

# HOUSEPLANTS AND AIR QUALITY

Houseplants improve air quality in our homes by absorbing carbon dioxide and releasing oxygen, helping to achieve the right balance needed for breathing and optimal brain function. They can also help reduce the buildup of problematic toxins emitted by various household products.

In our daily lives, we use numerous products that can leave residues in the air. Aerosols, cleaning products, gas stoves, open fires, paint, and cigarette smoke are just some of the sources of airborne toxins such as formaldehyde, benzene, xylene, and ammonia. Without proper ventilation, these toxins can build up in our homes and may cause headaches, nausea, low motivation, and irritation of the eyes and throat. They can also exacerbate skin problems and breathing difficulties such as asthma.



## Get the air moving

Plants such as the pink quill (*pp.132–133*) will appreciate an open window to boost airflow.

Plants can act as filters for these pollutants and help neutralize them. A pioneering study in 1989 by NASA into the air-purifying abilities of certain houseplants has since been largely backed up by a 2007 Australian study, which found that keeping three or more large plants per person in an office can detoxify the air of volatile organic compounds (VOCs) by up to 75 percent.

The average home would likely give different results, given the controlled environment in which these studies were carried out. But there are many things you can do to help your plants clean the air:

- The more plants, the greater the potential to clean the air. The exact number will depend on the size, maturity, and health of the plants, so just have as many as possible.
- Allow good air circulation by opening windows regularly and maximizing air movement (perhaps use a mechanized fan).
- Be sure your plants have the appropriate light, temperature, and humidity levels to optimize their health and efficacy.
- Help decrease the chemicals in your home by using low-emission products.